

Figure 8: A five-stage RISC pipeline

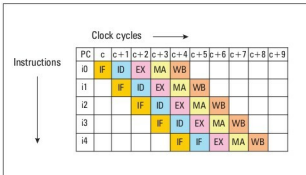


Figure 9: Execution of instructions in a five-stage RISC pipeline

not gold. In pipelining, it is not always as easy to obtain the throughput as we imagine it to be. We are often faced with a situation that prevents the next instruction in the instruction stream from being executed during its designated clock cycle. Such a situation is known as a *Pipeline Hazard*. Hazards reduce the performance speed gained by pipelining.

In my subsequent articles, I will discuss the concept of pipeline hazards, and how we can deal with them. I will also discuss how a developer can exploit this feature to increase the throughput. Till the next article, keep ‘meshin’! 

References and further reading

1. Sima, Fountain and Kacsuk -- *Advanced Computer Architectures -- A Design Space Approach*
2. http://www.cs.iastate.edu/~prabhu/Tutorial/PIPELINE/pipeline_title.html
3. http://en.wikipedia.org/wiki/Instruction_pipeline

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